

Problem Statement

The Indian Railway system operates with a fixed seating capacity for each train. However, a large number of passengers travel without reserved seats by purchasing general or platform tickets. This often leads to severe overcrowding inside coaches, especially during long-distance journeys.

Uncontrolled overcrowding creates unsafe travel conditions for passengers. Incidents such as theft, violence, medical emergencies, and passenger distress become more common in highly crowded compartments. Due to excessive crowd density, passengers also face difficulties in movement, seating, ventilation, and emergency evacuation.

The current railway ticketing infrastructure mainly relies on physical counter-based ticket sales and lacks real-time integration with a centralized passenger capacity management system. As a result, authorities are unable to effectively monitor or restrict the number of passengers boarding a train beyond its safe capacity.

Additionally, there is no automated monitoring system available to continuously track passenger density, detect suspicious or abnormal behavior, or generate real-time alerts for Railway Protection Force (RPF) officers and ticket checkers during active train journeys.

This absence of intelligent crowd control and safety monitoring creates a major challenge in ensuring passenger security, efficient travel management, and emergency response within the Indian Railway network.

Therefore, there is a need for an intelligent, automated, and multimodal railway monitoring system capable of managing passenger capacity, analyzing crowd conditions, detecting abnormal activities, and providing real-time alerts to railway authorities.